MIL-D-13670C (PA)
22 April 1975
SUPERSEDING
MIL-D-13670B (MU)
17 August 1964 and
MIL-D-46260A (MU)
17 August 1964
(see 6.6)

MILITARY SPECIFICATION

DETONATOR, STAB, M63
PARTS FOR
LOADING, ASSEMBLING AND PACKING

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers parts for, loading, assembling and packing for one type of detonator designated as Detonator, Stab, M63.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation - Packaging, Methods of
*MIL-A-48078- Ammunition, Standard Quality Assurance Provisions, General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

*(For meaning of asterisk (*) see 6.5)

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DRAWINGS

U.S. ARMY PICATINNY ARSENAL

*PRODUCT AND PACKING DRAWINGS

8798730 - Detonator, M63, Assembly
9258182 - Box, Set-up, Packing, Ammunition, Inner for Small Explosive Components
9258183 - Box, Set-up, Packing, Ammunition, Outer for Small Explosive Components
9258184 - Box, Packing, Ammunition for Small Explosive Components.
9297866 - Disc, Detonator
9297867 - Disc, Detonator Closing
9297868 - Cup Detonator

*INSPECTION EQUIPMENT DRAWINGS

8816014 - Ball, Steel
8816367 - Firing Pin Guide
8816368 - Disc
9200012 - Pin, Firing
9200390 - Firing Assembly
9200459 - Sleeve
9200602 - Bushing

(Copies of specifications, standards and drawings required by Contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Detonator. - The detonators shall comply with all requirements specified on Drawing (dwg. 8798730, and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide. - The moisture content of the lead azide used in loading the detonators shall not exceed 0.50 percent.

3.4 Moisture content of primer mixture. - The moisture content of the primer mixture used in loading the detonators shall not exceed 0.30 percent.

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3.5 Functioning

3.5.1 Output. - The detonators shall function, on the first attempt under an applied energy of 7.76 inch ounces, and shall cause a 0.16 inch diameter minimum (min), perforation in a .131 inch min. thick lead disc.

3.5.2 Waterproofness. - The detonators shall be waterproof as evidenced by normal functioning, as specified in 3.5.1 after being immersed in water maintained at 75 degrees Fahrenheit (F.) plus or minus 5 degrees F., at a depth of 2 inches minimum for not less than 48 hours.

*3.6 Workmanship. - All parts shall be fabricated and loaded in a thorough, workmanlike manner, and all manufacturing, processing and assembly operations shall be correctly performed. The parts shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. (All required packing shall be dry).

3.7 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, theprovisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. - The following types of inspection shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First Article Inspection

4.3.1 Submission. - The Contractor shall submit a first article sample (see 6.1b) as designated by the Contracting Officer. The first article sample shall consist of 50 complete sets of parts and sub-assemblies (i.e., 50 each of every component and sub-assembly and 365 detonator assemblies. The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

4.3.2 Inspections to be performed. - See MIL-A-48078 and paras. 4.4.2.1 thru 4.4.2.4, testing per para. 4.4.3, and Table I specified herein.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*	Detonator, Stab, M63, Inert Parts and Complete Assemblies.				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inert Parts (Dwg. 9297866, 9297867 and 9297868) Examination for Defects	50(a)		3.1	4.4.2.1 to 4.4.2.3
	Detonators (Complete Assemblies) (Dwg. 8798730) Examination for Defects	50(b,c)		3.2	4.4.2.4
	Tests Functioning Output	315		3.5.1	4.5.4.1
	Waterproofness	50		3.5.2	4.5.4.2
	NOTES: a) 50 each and every inert component b) Randomly selected from the 365 to be functioned. c) Inspect entire functioning sample (365) for Critical Defect.				

NOTES:
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*4.3.3 Rejection. -(See MIL-A-48078 (PA))

4.4 Quality Conformance Inspection

*4.4.1 Inspection Lot Formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Lot numbering, as required, shall be in accordance with MIL-STD-1168. In addition, inspection lots of Detonator Assemblies shall contain:

a. Parts of one lot interfix number from one manufacturer.
b. Primer mixture from one batch or a blending of batches which results in homogeneity of all primer mixture used in the loading of a lot.
c. Lead azide from one batch or a blending of batches which results in homogeneity of all explosives used in loading a lot.
d. RDX from one batch or a blending of batches which results in homogeneity of all explosives used in loading a lot.

4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.1	Disc, Detonator, Closing and Disc, Detonator				9297866 & 9297867
					NEXT HIGHER ASSEMBLY 8798730
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major B:					
131.	Thickness		0.40%	3.2	Gage
132.	Diameter		0.40%	3.2	Gage
Minor:					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-13670C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.2	Cup, Detonator				9297868
					NEXT HIGHER ASSEMBLY
					8798730
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup, cracked, split or cut		100.0%	3.2	Visual
Major B					
131	Diameter of flash hole, min		0.40%	3.2	Gage
132	Thickness through bottom, min.		0.40%	3.2	Gage
133	Concentricity of flash hole with outside diameter		0.40%	3.2	Gage
Minor					
201	Inside diameter		0.65%	3.2	Gage
202	Outside diameter		0.65%	3.2	Gage
203	Total length		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-13670C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8798730
*4.4.2.3	Assembly, Prior to Loading and Closing				NEXT HIGHER ASSEMBLY 8798730
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Coating on interior of cup missing or inadequate		100%	3.2	Visual
Major A&B	None defined				
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8798730
*4.4.2.4	Assembly - Detonator M63				NEXT HIGHER ASSEMBLY 8798730
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Cup cracked, split, or cut		100%	3.2	Visual
Major B:					
131.	Length		0.40%	3.2	Gage
132.	Diameter		0.40%	3.2	Gage
133.	Disc loose or punctured (either end) or wrinkled to such a degree that the contents are exposed		0.40%	3.2	Visual/Manual
134.	Crimp not full 360 degrees		0.40%	3.2	Visual
135.	Color of lacquer incorrect		0.40%	3.2	Visual
136.	Lacquer seal missing or inadequate		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9258182
*4.4.2.5	Box, Set-up, Inner, after Loading and Prior to Closing Slide Cover				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Filler and cushion assembly missing		100%	3.2	Visual
2.	Filler and cushion assembly improperly assembled		100%	3.2	Visual
3.	Stiffner missing		100%	3.2	Visual
4.	Slide cover assembled reversed		100%	3.2	Visual
Major A:					
101.	Release paper missing		0.40%	3.2	Visual
Minor:					
201.	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202.	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.6	Box, Set-up, Outer, with Inner Set-up Boxes Assembled and Prior to Assembling Top and Sealing Strip				9258183
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Red surfaces on inner set-up boxes do not alternate on same side of outer set-up box		100%	3.2	Visual
2.	Stiffener not facing one direction		100%	3.2	Visual
Major A-B:	None defined				
Minor:					
201.	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.7	Sealed packed barrier bag, prior to Packing in Box				9258183
					NEXT HIGHER ASSEMBLY 9258183
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Waterproofness of barrier bag destroyed by rupture, puncture, or separation of heat seal		0.40%	3.2	Visual
102.	Evidence of crushed box		0.40%	3.2	Visual
	NOTE: Package should be opended if there is evidence of crushing.				
Minor:					
201.	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied.		0.65%	3.2	Visual
202.	Marking of bag, misleading or unidentifiable		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.8	Box, Packing, Ammunition Prior to Assembling Cover and Strapping				9258184
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Bags containing set-up boxes damaged		0.40%	3.2	Visual
102.	Cushioning material incomplete or missing from around outer set-up boxes		0.40%	3.2	Visual
103.	Marking missing, misleading or unidentifiable		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION MIL-D-13670C (PA)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.9	Box, Packing, Ammunition				9258184
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Strapping missing, broken, loose or improperly fastened		0.40%	3.2	Visual/Manual
102.	Box damaged		0.40%	3.2	Visual
103.	Marking missing, incomplete or illegible		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

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4.4.3 Testing

4.4.3.1 Moisture content of lead azide (see 3.3). - Major A defect - The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample shall be selected and tested as specified in 4.5.1. If the moisture content is in excess of the percentage specified and if loading has not begun, the lot of lead azide shall be rejected. If loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded with questionable lead azide shall be rejected.

4.4.3.2 Primer mixture

4.4.3.2.1 Moisture content (see 3.4). - Major A defect. The moisture content of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample shall be selected and tested as specified in 4.5.1. If the moisture content is in excess of the percentage specified and if loading has not begun, the batch or blend of batches of the primer mixture shall be rejected. If loading has begun, action on the unloaded primer shall be as specified above and all detonators loaded with the questionable primer mixture shall be rejected.

4.4.3.2.2 Composition (see dwg. 8798730) Major A defect. - The composition of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample, shall be selected and tested as specified in 4.5.2. If the composition of the primer mixturefails to conform to the requirements specified on the applicable drawing and if loading has not begun, the batch or blend of batches of primer mixture shall be rejected. If loading has begun, detonators loaded with the questionable primer mixture shall be rejected.

4.4.3.3 Binder-lubricant content of RDX (see dwg. 8798730) - Major A defect - The binder-lubricant content of the RDX shall be determined for each lot at the time of loading. A sample of five grams shall be selected from each lot and tested as specified in 4.5.3. If the binder-lubricant content is in excess of the percentage specified and if loading has not begun, the lot of RDX shall be rejected until corrective action has been accomplished as verified by repeating the binder-lubricant determination specified in 4.5.3. If loading of the detonators has begun, the determination shall be made on the unloaded RDX. If the binder-

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lubricant content of the unloaded RDX is in excess of the percentage specified, the unloaded RDX shall be rejected until corrective action has been accomplished and all detonators loaded with RDX from the rejected lot shall be rejected.

4.4.3.4 Functioning

4.4.3.4.1 Output (see 3.5.1). - Major A defect. - A sample of 315 detonators shall be selected at random from each lot. If 3 or more detonators fail to comply with the requirement specified the lot shall be rejected. If one or two defects are found a second sample of 315 shall be selected at random and tested. If, in the combined first and second sample a total of 4 or more defects are found the lot shall be rejected. The test shall be performed as specified in 4.5.4.1. (Destructive Test)

4.4.3.4.2 Waterproofness (see 3.5.2). - Major A defect - A sample of 25 detonators shall be selected at random from each lot. If one or more detonators fail to comply with the requirement specified the lot shall be rejected. The test shall be performed as specified in 4.5.4.2. (Destructive Test)

4.4.3.5 Heat seal test of sealed bag (see dwg. 9258183). - Major A defect - Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-P-116. (Destructive Test).

*4.4.4 Inspection equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1 and the test method paragraphs (see 4.5). The Contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and Section 6.2 herein.

*4.5 Test methods and procedures

*4.5.1 Moisture content of lead azide and primer mixture. - The moisture content of the lead azide and primer mixture shall be determined in accordance with MIL-STD-650, Method 101.6 except that a two (2) gram sample shall be used.

*4.5.2 Composition of primer mixture. - The composition of the primer mixture shall be determined by performing the following tests in the sequence listed.

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*4.5.2.1 Barium nitrate. - An accurately weighed portion of approximately 0.5 gram of primer mixture shall be transferred to a tared medium-porosity, fritted glass crucible. Three ml of lead azide saturated distilled water shall be added at a temperature of 50 plus or minus 2 degrees C., agitated for 1 minute, and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five times, making a total of six 3-ml. extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, aspirated until the odor of alcohol is no longer detectable, dried in an oven maintained at 60 plus or minus 5 degrees C., for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barium nitrate, percent} = \frac{A-B}{W}$$

Where:

A = weight of crucible and sample
B = weight of the crucible and sample after extractions with lead azide saturated water
W = weight of sample

4.5.2.2 Lead styphnate

4.5.2.2.1 Basic lead styphnate (when applicable). - The residue obtained in the determination of barium nitrate shall be extracted with six-3ml. portions of saturated ammonium acetate solution at about 25 degrees C., (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250-ml. calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2-ml. aliquot of this solution shall be transferred to a 50-ml. calibrated volumetric flask and diluted to the mark with distilled water. A portion of this selection shall be transferred to a corex glass spectrophotometric cell having a width of approximately 1 centimeter (cm.), and its absorbance density determined as a wave length of 410 millimicrons. As reference, a similar cell containing ammonium acetate solution of the same concentration as the sample

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solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the absorbance density shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbance density of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated as follows:

Basic lead styphnate, percent = F(A-B) divided by W(1-.01M)D

Where:

A = absorbance density of basic lead styphnate solution
B = absorbance density of ammonium acetate solution
D = width of corex cell, cm.
M = moisture content of sample obtained from 4.5.1 in percent
W = weight of sample
F = Factor to be determined for each instrument on the basis of a sample of lead styphnate that is suitable for use as a standard.

*4.5.2.2.2 Determination of normal lead styphnate (when applicable). Use method 408.1 of MIL-STD-650.

*4.5.2.3 Tetracene. - The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol aspirated until the odor of alcohol is gone, then dried in an oven maintained at 60 plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125-ml. beaker with a stream of water, 25 ml. of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible and the residue in the crucible washed three times with boiling water, and finally with alcohol, then aspirate until the odor of alcohol is no longer detectable. The crucible and contents shall be dried in an oven maintained at 60 plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

Tetracene, percent = (C-D) 100 divided by W (1-.01M)

Where:

C = weight of crucible and residue after extraction with saturated ammonium acetate solution

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D = weight of crucible and residue after treating with hot water
W = weight of the sample
M = moisture content of sample obtained from 4.5.1 in percent

*4.5.2.4 Antimony sulfide. - The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

Antimony sulfide, percent = (D-E) 100 divided by W(1.-.01M)

Where:
D = weight of crucible and sample after treating with hot water
E = weight of crucible
W = weight of sample
M = moisture content of sample obtained in 4.5.1 in percent.

4.5.2.5 Lead azide. - The lead azide content in the sample on a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, basic lead styphnate, tetracene and antimony sulfide.

4.5.3 Determination of binder-lubricant content of RDX. - The calcium resinate, graphite, and total binder-lubricant content shall be determined as follows:

4.5.3.1 Calcium resinate. - An accurately weighed portion of 5 grams of the sample shall be transferred to a tared medium porosity sintered-glass crucible. The sample shall be extracted with three 20 milliliter (ml) portions of chloroform which have been previously saturated with RDX, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up by stirring occasionally and the sides of the crucible washed down with each 20 ml. portion of chloroform to insure complete removal of the calcium resinate. The residue shall be aspirated until the odor of chloroform is no longer detectable; the crucible and contents shall be placed in a drying oven at 100-105 degrees Centigrade (°C.) for 1/2 hour, cooled in a desiccator and weighed. The above extraction shall be repeated, drying, and weighing with individual 20 ml. portions of chloroform until the loss in weight on subsequent extractions totals less than 0.003 grams. The loss in weight of the crucible plus contents shall be calculated to percent calcium resinate in the sample on a moisture-bree basis as follows:

$$\text{Percentage of Calcium Resinate} = \frac{100A}{W}$$

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Where:
A = loss in weight of the crucible plus contents
W = weight of the sample

4.5.3.2 Graphite. - The residue remaining after the calcium resinate has been removed shall be extracted with two 20 ml. portions of dimethyl formamide, each portion maintained in contact with the residue for approximately 5 minutes with intermittent stirring before applying suction. The crucible and contents shall be washed with a minimum of 3 ml. portions of acetone, each portion maintained in contact with the residue for approximately 1 minute stirring before applying suction, until the following qualitative test for RDX in the acetone washings is negative. The acetone filtrate shall be transferred to a 25 ml. beaker, evaporated to dryness on a steam bath and several drops of a solution of chromotropic acid added in 36N sulfuric acid. The formation of a reddish-brown color, as compared to a pink color obatined by a blank prepared in an identical manner indicates the presence of RDX. The residue shall be aspirated until the odor of acetone is no longer detectable; the crucible and contents shall be placed in a drying oven at 95 degrees C. plus or minus 2 degrees C. for 1/2 hour, cooled in a desiccator and weighed. The weight of the residue in the crucible shall be calculated to the percentage of graphite in the sample on a moisture-free basis as follows:

$$\text{Percentage of Graphite} = \frac{100A}{W}$$

Where:
A = increase in weight of filtering crucible
W = weight of sample

4.5.3.3 Total binder-lubricant . - The percentage of total binder-lubricant in the sample shall be calculated by adding the percentages of calcium resinate and graphite found, and substracting from 100.

4.5.4 Functioning

*4.5.4.1 Output. - The detonator shall be assembled to the applicable test fixture so that the firing pin will strike the sensitive end of the detonator. A steel ball weighing 1.94 ounces maximum shall be dropped on the firing pin to function the detonator from a height of 4 inches, maximum. The height of drop shall be measured as the height from the top of the firing

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pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective. Method of testing shall be in accordance with dwgs. 8816014, 8816367, 8816368, 9200012, 9200390, 9200459 and 9200602.

*4.5.4.2 Waterproofness . - The detonator shall be immersed under water as specified for the time specified. Within one hour after removal from the water the detonator shall be subjected to the output test specified in 4.5.4.1. Any detonator which fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective. Method of testing shall be in accordance with dwgs. specified in 4.5.4.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

*5.1.1 Level C. - Minimum Military Package. - The detonators shall be packaged and marked in accordance with dwg. 9258182 and dwg. 9258183.

*5.1.2 Level C .- Detonator parts - Packaging shall be in accordance with MIL-STD-1169 requirements.

5.2 Packing

*5.2.1 Level C. - Detonators shall be packed as specified on dwg. 9258184.

*5.2.2 Level C. - Detonator parts shall be packed in accordance with MIL-STD-1169 requirements.

*5.3 Marking. -- Detonator containers and bags shall be marked in accordance with instructions contained in dwgs. 9258182, 9258183 and 9258184. Detonator parts containers shall be marked in accordance with MIL-STD-1169 requirements.

6. NOTES

*6.1 Ordering Data. -- Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples
c. Provisions of MIL-A-48078.

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*6.2 Submission of Inspection Equipment Designs for Approval (see MIL-A-48078). Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T Dover, New Jersey 07801.

*6.3 Distribution of ammunition data cards . -- Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey-, 07801.

6.4 Check test for possible deterioration. - The following test is designed to prevent the loading of detonators which have deteriorated, into a major ammunition item. The test shall be performed on detonators from each lot which has been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in that contract. The Contracting Officer will arrange for the contractor to be reimbursed for expenses incurred in the performance of the check test for possible deterioration.

*6.4.1 Output. - A sample of 315 detonators shall be selected at random from each lot. If 4 or more detonators fail to comply with the requirement specified, the lot shall be rejected. If more than one defective is found, a second sample of 315 shall be selected at random and tested. If, in the combined first and second sample, a total of 5 or more defectives are found, the lot shall be rejected. The test shall be performed as specified in 4.5.4.1.

*6.5 The margins of this specifcation are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previosu issue.

*6.6 Supersession Data . - This specification includes requirements of MIL-D-46260A (MU) dated 17 August 1964.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project No. 1375-A-134

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

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